

# IPv4 and IPv6 — Guided Notes

Unit 1.4.2 | CompTIA Network+ N10-009 Obj. 1.4 | N10-008 1.4

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

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Complete each question using the lesson reading. Write in the blank or answer in the space provided.

## Question 1

An IPv4 address is \_\_\_\_\_ bits long, written as \_\_\_\_\_ octets in dotted decimal notation. Each octet ranges from \_\_\_\_\_ to \_\_\_\_\_. The total number of possible IPv4 addresses is approximately \_\_\_\_\_.

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## Question 2

Every IPv4 address has two parts: a \_\_\_\_\_ portion and a \_\_\_\_\_ portion. The \_\_\_\_\_ determines where the split falls. A /24 mask means \_\_\_\_\_ bits are network and \_\_\_\_\_ bits are host.

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## Question 3

CIDR notation /30 leaves \_\_\_\_\_ bits for hosts, providing \_\_\_\_\_ total addresses and \_\_\_\_\_ usable addresses. This is typically used for \_\_\_\_\_.

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## Question 4

IPv6 addresses are \_\_\_\_\_ bits long, written as \_\_\_\_\_ groups of \_\_\_\_\_ hexadecimal digits separated by colons. Two shorthand rules are: (1) \_\_\_\_\_ and (2) \_\_\_\_\_.

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## Question 5

A standard IPv6 subnet assignment uses a \_\_\_\_\_ prefix, giving \_\_\_\_\_ bits for the host identifier. Devices can generate their own host portion automatically using \_\_\_\_\_ (SLAAC) without needing a DHCP server.

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## Question 6

IPv6 has NO \_\_\_\_\_ addresses. Instead it uses three communication types:

- \_\_\_\_\_ — one to one
- \_\_\_\_\_ — one to many (replaces broadcast functions)

\_\_\_\_\_ — one to nearest

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### Question 7

Match each IPv6 address prefix to its type:

fe80::/10 → \_\_\_\_\_

2000::/3 → \_\_\_\_\_

ff00::/8 → \_\_\_\_\_

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### Question 8 — Real World Application

REAL WORLD: You run ipconfig on a Windows workstation and see both a 192.168.1.45 address and a 2001:db8::42 address on the same adapter. What is this configuration called, and what does it mean for how the machine connects to the internet?

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*Download the answer key from the class server:*

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wget http://[SERVER]/resources/network-engineer/1-4-2-ipv4-and-ipv6/1-4-2-ipv4-and-ipv6-answer-key.pdf
```